Untitled1

September 9, 2026

[4]:
```python
# Exercise 1 : Student Information using Dictionary
my_dict = {
    "Name": "ABC",
    "Roll No": 34,
    "Branch": "Computer Engineering",
    "Division": 12,
    "Semester": 1
}
print("Original Student Information:")
print(my_dict)

my_dict["College"] = "MIT-WPU"

print("\nAfter Adding College:")
print(my_dict)

del my_dict["Division"]

print("\nAfter Deleting Division:")
print(my_dict)

print("\nFinal Student Information:")
for key, value in my_dict.items():
    print(key, ":", value)
```

```
Original Student Information:
{'Name': 'ABC', 'Roll No': 34, 'Branch': 'Computer Engineering', 'Division': 12,
'Semester': 1}

After Adding College:
{'Name': 'ABC', 'Roll No': 34, 'Branch': 'Computer Engineering', 'Division': 12,
'Semester': 1, 'College': 'MIT-WPU'}

After Deleting Division:
{'Name': 'ABC', 'Roll No': 34, 'Branch': 'Computer Engineering', 'Semester': 1,
'College': 'MIT-WPU'}

Final Student Information:
```

```
Name : ABC
Roll No : 34
Branch : Computer Engineering
Semester : 1
College : MIT-WPU
```

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```python
# Exercise 2: Convert two lists into a dictionary

list1 = ['name', 'division', 'rollno']
list2 = ['ABC', 12, 1098]
my_dict = dict(zip(list1, list2))
print("Dictionary:")
print(my_dict)
```

```
Dictionary:
{'name': 'ABC', 'division': 12, 'rollno': 1098}
```

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```python
# Exercise 3: Sort Dictionary Elements

my_dict = {
    'name': 'ABC',
    'division': 12,
    'rollno': 1098,
    'marks': [65, 87, 93, 95]
}
print("Original Dictionary:")
print(my_dict)

sorted_dict = sorted(my_dict)

print("\nSorted Dictionary Keys:")
print(sorted_dict)
```

```
Original Dictionary:
{'name': 'ABC', 'division': 12, 'rollno': 1098, 'marks': [65, 87, 93, 95]}

Sorted Dictionary Keys:
['division', 'marks', 'name', 'rollno']
```

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```python
# Exercise 4: Convery Dictionary Keys and Values into Lists

my_dict = {
    'name': 'ABC',
    'division': 12,
    'rollno': 1098
}
keys_list = list(my_dict.keys())
values_list = list(my_dict.values())
```

```
print("Dictionary:", my_dict)
print("Keys List:", keys_list)
print("Values List:", values_list)
```

```
Dictionary: {'name': 'ABC', 'division': 12, 'rollno': 1098}
Keys List: ['name', 'division', 'rollno']
Values List: ['ABC', 12, 1098]
```

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# exercise 5: Find Mean of Dictionary Values

my_dict = {
    'marks1': 23,
    'marks2': 35,
    'marks3': 13,
    'marks4': 26,
    'marks5': 39
}
values = my_dict.values()
mean = sum(values) / len(my_dict)
print("Dictionary:", my_dict)
print("Mean of all values:", mean)
```

```
Dictionary: {'marks1': 23, 'marks2': 35, 'marks3': 13, 'marks4': 26, 'marks5':
39}
Mean of all values: 27.2
```

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```
# Exercise 6: Dictionary Operations

my_dict = {
    'name': ['Yash', 'Neel', 'Dev'],
    'rollno': [10, 11, 12],
    'marks': [78, 56, 98]
}
print("Name:", my_dict['name'][2])
print("Rollno:", my_dict['rollno'][1])
print("Greatest Marks:", max(my_dict['marks']))
```

```
Name: Dev
Rollno: 11
Greatest Marks: 98
```

[15]:
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# Exercise 7: Frequency of Characters in a String

string = input("Enter a string:")
frequency = {}
for char in string:
    if char in frequency:
```

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        frequency[char] += 1
    else:
        frequency[char] = 1
print("Frequency of Characters:")
print(frequency)
```

```
Enter a string: hello world

Frequency of Characters:
{'h': 1, 'e': 1, 'l': 3, 'o': 2, ' ': 1, 'w': 1, 'r': 1, 'd': 1}
```

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# Exercise 8: Quotient and Remainder

numerator = int(input("Enter numerator:"))
denominator  = int(input("Enter denominator:"))

quotient = numerator // denominator
remainder = numerator % denominator

result = (quotient, remainder)

print("Quotient and Remainder:",
result)
```

```
Enter numerator: 17
Enter denominator: 5

Quotient and Remainder: (3, 2)
```

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# Exercise 9: Find Number of Days Between Two Dates

from datetime import date

d1 = int(input("Enter first day:"))
m1 = int(input("Enter first month:"))
y1 = int(input("Enter first year:"))
date1_tuple = (d1, m1, y1)
date1 = date(y1, m1, d1)

d2 = int(input("Enter second day:"))
m2 = int(input("Enter second month:"))
y2 = int(input("Enter second year:"))
date2_tuple = (d2, m2, y2)
date2 = date(y2, m2, d2)

difference = abs((date2 -
date1).days)
print("\nFirst Date Tuple:", date1_tuple)
```

```
print("Second Date Tuple:", date2_tuple)
print("Number of days between the two dates:", difference)
```

```
Enter first day: 4
Enter first month: 11
Enter first year: 2005
Enter second day: 6
Enter second month: 9
Enter second year: 2008


First Date Tuple: (4, 11, 2005)
Second Date Tuple: (6, 9, 2008)
Number of days between the two dates: 1037
```

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